```
!pip install pgmpy pandas
```

```
Collecting pgmpy
  Downloading pgmpy-1.1.0-py3-none-any.whl.metadata (13 kB)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
Requirement already satisfied: huggingface_hub>=0.23 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (1.7.1)
Requirement already satisfied: networkx>=3.0 in /usr/local/lib/python3.12/dist-packages (from pgmpy) (3.6.1)
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Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
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Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface_hub>=0.23->pgmpy) (
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Installing collected packages: scikit-base, pgmpy
Successfully installed pgmpy-1.1.0 scikit-base-0.13.1
```

```
from pgmpy.models import DiscreteBayesianNetwork
from pgmpy.estimators import MaximumLikelihoodEstimator
from pgmpy.inference import VariableElimination
import pandas as pd

# ----------------------------------------
# Step 1: Create sample dataset
# ----------------------------------------
data = pd.DataFrame({
    'Income': ['High', 'Low', 'Medium', 'Low', 'High', 'Medium', 'Low', 'High'],
    'CreditScore': ['Good', 'Poor', 'Good', 'Poor', 'Good', 'Good', 'Poor', 'Good'],
    'LoanAmount': ['Low', 'High', 'Medium', 'High', 'Low', 'Medium', 'High', 'Low'],
    'Default': ['No', 'Yes', 'No', 'Yes', 'No', 'No', 'Yes', 'No']
})

print("Dataset:\n", data, "\n")

# ----------------------------------------
# Step 2: Define Bayesian Network structure
# ----------------------------------------
model = DiscreteBayesianNetwork([
    ('Income', 'LoanAmount'),
    ('CreditScore', 'Default'),
    ('LoanAmount', 'Default')
])

# ----------------------------------------
# Step 3: Train model using MLE
```

```
# ----------------------------------------
model.fit(data, estimator=MaximumLikelihoodEstimator)

# ----------------------------------------
# Step 4: Print CPDs
# ----------------------------------------
print("Conditional Probability Distributions (CPDs):\n")
for cpd in model.get_cpds():
    print(cpd)
    print()

# ----------------------------------------
# Step 5: Perform inference
# ----------------------------------------
inference = VariableElimination(model)

# Query 1: P(Default | Income=Low, CreditScore=Poor)
result1 = inference.query(
    variables=['Default'],
    evidence={'Income': 'Low', 'CreditScore': 'Poor'}
)

print("P(Default | Income=Low, CreditScore=Poor):\n")
print(result1)

# ----------------------------------------
# Additional Queries
# ----------------------------------------

# Query 2: P(Default | Income=High, CreditScore=Good)
result2 = inference.query(
    variables=['Default'],
    evidence={'Income': 'High', 'CreditScore': 'Good'}
)

print("\nP(Default | Income=High, CreditScore=Good):\n")
print(result2)

# Query 3: P(LoanAmount | Income=Low)
result3 = inference.query(
    variables=['LoanAmount'],
    evidence={'Income': 'Low'}
)

print("\nP(LoanAmount | Income=Low):\n")
print(result3)
```

```
7    High        Good        Low      No

Conditional Probability Distributions (CPDs):

+----------------+-------+
| Income(High)   | 0.375 |
+----------------+-------+
| Income(Low)    | 0.375 |
+----------------+-------+
| Income(Medium) | 0.25  |
+----------------+-------+

+--------------------+--------------+-------------+----------------+
| Income             | Income(High) | Income(Low) | Income(Medium) |
+--------------------+--------------+-------------+----------------+
| LoanAmount(High)   | 0.0          | 1.0         | 0.0            |
+--------------------+--------------+-------------+----------------+
| LoanAmount(Low)    | 1.0          | 0.0         | 0.0            |
+--------------------+--------------+-------------+----------------+
| LoanAmount(Medium) | 0.0          | 0.0         | 1.0            |
+--------------------+--------------+-------------+----------------+

| Default(No)  | 0.5                 |      | 0.5                 |
```

```
| Default(No)  | 0.5                | ... | 0.5                |
+--------------+--------------------+-----+--------------------+
| Default(Yes) | 0.5                | ... | 0.5                |
+--------------+--------------------+-----+--------------------+

P(Default | Income=Low, CreditScore=Poor):

+--------------+----------------+
| Default      |   phi(Default) |
+==============+================+
| Default(No)  |         0.0000 |
+--------------+----------------+
| Default(Yes) |         1.0000 |
+--------------+----------------+

P(Default | Income=High, CreditScore=Good):

+--------------+----------------+
| Default      |   phi(Default) |
+==============+================+
| Default(No)  |         1.0000 |
+--------------+----------------+
| Default(Yes) |         0.0000 |
+--------------+----------------+
```